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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSESDV200 –

ELECTRO ACOUSTICS Answer Keys

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least **one question** from each course outcome to enhance their overall outcome attainability.

Part A

Very short answer Answer any **8** questions
Each question carries **2** marks

1. A sine wave inside a circle (~).
2. Left channel is **White** & Right channel is **Red**.
3. It is caused by the phase difference between the front and rear of the diaphragm in pressure-gradient microphones (like cardioids). As the source gets closer, the low-frequency boost increases.
4. Also known as supra-aural headphones, these have pads that press directly against the lobes of the ears rather than surrounding them.
5. A small omnidirectional condenser mic capsules mounted flush in a housing that places it very close to a flat surface (boundary) to eliminate phase interference from reflections.
6. An electrical device that converts alternating current (AC), which periodically reverses direction, to direct current (DC), which flows in only one direction.
7. One major factor is **capacitance**; high capacitance in long cable runs can bleed off high-frequency signals.
8. **Microphone Sensitivity Unit:** Typically expressed in **mV/Pa** (millivolts per Pascal) or **dBV/Pa**.
9. High efficiency - they can produce high sound pressure levels with relatively low amplifier power – convert high pressure to low pressure – make airflow smooth.
10. A metric used to quantify the ability of a balanced device (like a preamp) to reject input signals common to both input leads (noise/interference) while passing the differential signal.

Part B

Short answer Answer any **5** questions,
Each question carries **6** marks

11. **Stereophonic microphones** are actually microphone capsules with **electronically separate systems** housed in a **single casing** or **separate casings**- Configured one of three ways- Lower and upper elements stationary- one element stationary and the other rotates 180 to 360 degrees – Both elements rotate.

12. Crossovers divide the signal so that low frequencies go to woofers and high frequencies go to tweeters, preventing distortion and damage.
13. **Line Cables-** carry low-voltage, high-impedance signals. They are shielded to protect against EMI/RFI interference- **Speaker Cables-** carry high-voltage, low-impedance signals. They are unshielded and use thicker gauge wire to handle high current without significant power loss.
14. **Occurs** when an amplifier is pushed beyond its voltage limits, "cutting off" the peaks and troughs of the waveform- The sine wave peaks become flattened, resembling a square wave- Minimise by lowering input gain, using a limiter, or using an amplifier with higher headroom.
15. Operates on the principle of a variable capacitor - a thin **diaphragm** and a solid **backplate** act as the two plates of a capacitor. Sound waves move the diaphragm, changing the capacitance and creating an electrical signal. It requires phantom power (+48V).
16. It describes how a speaker radiates sound into a space relative to its axis. At low frequencies, speakers are usually omnidirectional - at high frequencies, they become more directional.
17. This relationship is defined by **Ohm's Law** (V=IR), where Voltage (V) is the pressure, Current (I) is the flow of electric charge, and Resistance (R) is the opposition to that flow.

Part C

Essay. Answer any **2** questions, each question carries **12** marks

18. Advantages:
- High bandwidth, low distortion
- Immune to electromagnetic interference
- Low signal loss, long-distance transmission
- Electrical isolation prevents ground loops
- Limitations:
- Higher cost
- Fragile, prone to breakage
- Requires special transmitters/receivers
- Difficult to repair/splice.

19. Primarily refers to Shotgun Microphones- They use an interference tube to cancel out sound arriving from the sides through destructive interference while maintaining high sensitivity for sound directly in front. Essential for film sets and field recording- also explain about Super and hyper cardioid.
20. **Baffles:** A board used to mount drivers, preventing the rear sound wave (which is 180 degrees out of phase) from cancelling the front wave. **Enclosures** include **Sealed (Acoustic Suspension)** for tight bass, and **Ported (Bass Reflex),** which uses a tuned vent to increase low-frequency efficiency.